

3 (a) Define power.

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 [1]

(b) An electric car is powered by a motor. The car is travelling at a constant speed of 35 ms^{-1} along a straight horizontal road, as shown in Fig. 3.1.

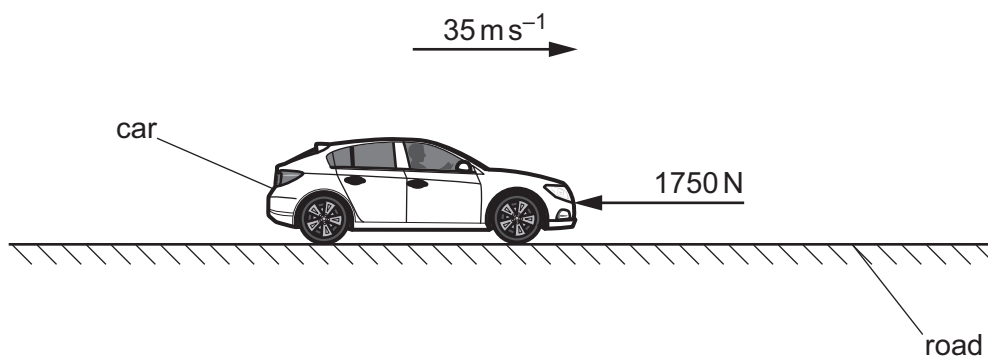


Fig. 3.1

There is a total resistive force of 1750 N acting on the car.

(i) Calculate the power transmitted to the wheels of the car by the motor.

power =W [2]

(ii) Calculate the useful work done by the motor when the car travels a distance of 17 km .

work done =J [2]

